ECOMMERCE CUSTOMER SALES ANALYSIS

POWER QUERY+SQL INTEGRATED PROJECT

ABSTRACT

This project involves analyzing ecommerce customer sales data using SQL. The raw dataset is first cleaned and standardized to remove inconsistencies, null values, and duplicates. It is then normalized into multiple related tables following the principles of database design up to the third normal form (3NF). Through structured SQL queries, various business insights are extracted — including revenue trends, top-performing products, customer behavior, and campaign effectiveness. The project demonstrates how SQL can be used to transform raw data into valuable, decision-making insights.

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Problem Statement:

The ecommerce platform is collecting large volumes of sales and customer data, but lacks structured insights into revenue trends, product performance, and customer behavior. Without proper data structuring, cleaning, and analysis, it's difficult to identify top-selling products, high-value customers, or campaign effectiveness — limiting data-driven business decisions.

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1. What's Happening?

- **Unstructured and messy data with missing values, inconsistencies, and duplicates**
- **No clear understanding of what's driving sales, revenue, or customer value**
- **No way to perform advanced analysis or build dashboards directly from raw data**

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2. Solution Goal:

- **Clean and normalize raw ecommerce sales data into structured SQL tables**
- **Perform analytical SQL queries to uncover revenue trends, top customers, and product performance**
- **Build a Power BI dashboard for actionable business insights and visualization**
- **Enable ongoing reporting and better decision-making using SQL + BI tools**

Data Description: ***(Title: Ecommerce Customer Sales Dataset)***

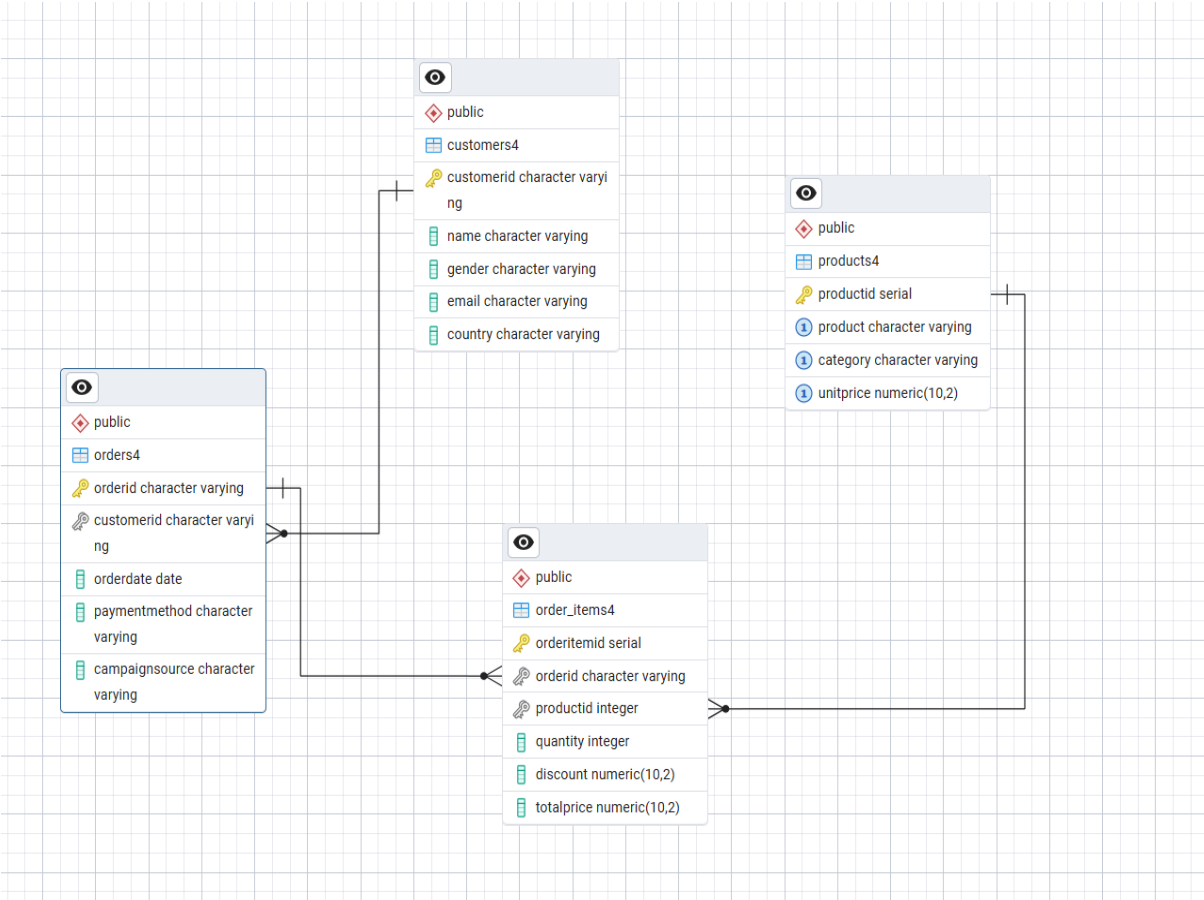
- **Sample size: ~30,000 Rows, 15 Columns**
- **Data Types: Mixed (Text, Numeric, Date, Categorical)**

- **Key Fields:**
    - **OrderID, CustomerID, Name, Gender, Country,**
    - **Product, Category, Quantity, UnitPrice, Discount, TotalPrice,**
    - **OrderDate, PaymentMethod, Email, CampaignSource**

Data Cleaning Steps:

- **Remove unnecessary/irrelevant columns**
- **Handle missing values (e.g., gender, prices)**
- **Remove duplicates and fix inconsistent formats**
- **Standardize categorical values (e.g., gender: “f”, “female”)**
- **Recalculate corrupted TotalPrice values**
- **Normalize data into separate SQL tables (3NF): customers, products, orders, order_items**

ER DIAGRAM


public
customers4
customerid character varying
name character varying
gender character varying
email character varying
country character varying
public
products4
productid serial
product character varying
category character varying
unitprice numeric(10,2)
public
orders4
orderid character varying
customerid character varying
orderdate date
paymentmethod character varying
campaignsource character varying
public
order_items4
orderitemid serial
orderid character varying
productid integer
quantity integer
discount numeric(10,2)
totalprice numeric(10,2)

SECTION 1: OVERALL BUSINESS METRICS

1. Total Revenue

Syntax:

```
SELECT SUM(TotalPrice) AS total_revenue
FROM order_items4;
```

	total_revenue numeric
1	1329367470.00

Insight:

Total revenue is the sum of all sales across customers, products, and time.

What this tells us:

It's the most important metric to understand how much money the business earned overall.

Helps track business growth and success over time.

2.Total Orders

Syntax:

```
SELECT COUNT(*) AS total_orders
FROM orders4;
```

	total_orders bigint
1	50000

Insight:

Total orders mean how many times customers have placed an order.

What this tells us:

It shows how active the sales are.

More orders usually mean more customer demand.

3. Total Customers

```
SELECT COUNT(DISTINCT CustomerID) AS total_customers
FROM customers4;
```

	total_customers bigint
1	1000

Insight:

Total customers means how many different people bought something.

What this tells us:

It shows how big the customer base is and how many people the business is reaching.

SECTION 2: PRODUCT PERFORMANCE

1. Top 5 Best-Selling Products (by Quantity)

```sql
SELECT p.Product, SUM(oi.Quantity) AS total_units_sold
FROM order_items4 oi
JOIN products4 p ON oi.ProductID = p.ProductID
GROUP BY p.Product
ORDER BY total_units_sold DESC
LIMIT 5;
```

	product character varying	total_units_sold bigint
1	Laptop	15372
2	Handbag	15166
3	T-shirt	15115
4	Headphones	15109
5	Dress	15042

Insight:

Top-selling products are the ones customers bought the most.

What this tells us:

Shows which products are popular and in high demand.

2 . Top 5 Products by Revenue

```sql
SELECT p.Product, SUM(oi.TotalPrice) AS revenue
FROM order_items4 oi
JOIN products4 p ON oi.ProductID = p.ProductID
GROUP BY p.Product
ORDER BY revenue DESC
LIMIT 5;
```

	product character varying	revenue numeric
1	Laptop	761772000.00
2	Mobile Phone	333163750.00
3	Smartwatch	53414640.00
4	Sneakers	45581375.00
5	Handbag	40966950.00

Insight:

These products made the highest total sales amount.

What this tells us:

They bring in the most money, even if they're not the top in quantity sold.

SECTION 3: CUSTOMER BEHAVIOR

1. Top 5 Customers by Spending

```sql
SELECT c.Name, SUM(oi.TotalPrice) AS total_spent
FROM customers4 c
JOIN orders4 o ON c.CustomerID = o.CustomerID
JOIN order_items4 oi ON o.OrderID = oi.OrderID
GROUP BY c.Name
ORDER BY total_spent DESC
LIMIT 5;
```

	name character varying	total_spent numeric
1	Christopher Smith	3265805.00
2	Nicholas Roman	2994080.00
3	Christopher Hahn	2848850.00
4	Logan Warren	2739040.00
5	Christopher Simpson	2694470.00

Insight:

These are the customers who spent the most money.

What this tells us:

They are our most valuable customers and can be given special offers or rewards.

2. Revenue by Country

```sql
SELECT c.Country, SUM(oi.TotalPrice) AS revenue
FROM customers4 c
JOIN orders4 o ON c.CustomerID = o.CustomerID
JOIN order_items4 oi ON o.OrderID = oi.OrderID
GROUP BY c.Country
ORDER BY revenue DESC;
```

	country character varying	revenue numeric
1	UK	292723650.00
2	Bangladesh	279474940.00
3	India	264134025.00
4	UAE	258577660.00
5	USA	234457195.00

Insight:

This shows how much money came from each country.

What this tells us:

Helps us see which countries bring the most sales and where to focus more.

3. Revenue by Gender

```sql
SELECT c.Gender, SUM(oi.TotalPrice) AS revenue
FROM customers4 c
JOIN orders4 o ON c.CustomerID = o.CustomerID
JOIN order_items4 oi ON o.OrderID = oi.OrderID
GROUP BY c.Gender;
```

	gender character varying	revenue numeric
1	M	639978360.00
2	F	689389110.00

Insight:

This shows how much male and female customers spent.

What this tells us:

Helps us see which group is buying more and if both should be targeted equally.

SECTION 4: MARKETING & PAYMENT INSIGHTS

1. Revenue by Payment Method

```
SELECT o.PaymentMethod, SUM(oi.TotalPrice) AS revenue
FROM orders4 o
JOIN order_items4 oi ON o.OrderID = oi.OrderID
GROUP BY o.PaymentMethod
ORDER BY revenue DESC;
```

	paymentmethod character varying	revenue numeric
1	UPI	340649575.00
2	Credit Card	335959710.00
3	Debit Card	331352075.00
4	Cash	321406110.00

Insight:

This shows how much money came from each payment method.

What this tells us:

Helps us understand how people prefer to pay and if we should improve or add more options.

2. Campaign Performance

```sql
SELECT o.CampaignSource, SUM(oi.TotalPrice) AS revenue
FROM orders4 o
JOIN order_items4 oi ON o.OrderID = oi.OrderID
GROUP BY o.CampaignSource
ORDER BY revenue DESC;
```

	campaignsource character varying	revenue numeric
1	Email Blast	271787835.00
2	Google Ads	271035680.00
3	Facebook	264559995.00
4	YouTube Ads	263435270.00
5	Instagram	258548690.00

Insight:

This shows how much revenue came from each marketing source like Google Ads or Email.

What this tells us:

Helps find out which campaigns are working best and bringing in

2.Month-over-Month Revenue Growth %

	month timestamp with time zone	revenue numeric	prev_month numeric	growth_percent numeric
1	2023-07-01 00:00:00+05:30	59281755.00	[null]	[null]
2	2023-08-01 00:00:00+05:30	55002260.00	59281755.00	-7.22
3	2023-09-01 00:00:00+05:30	52770670.00	55002260.00	-4.06
4	2023-10-01 00:00:00+05:30	55672825.00	52770670.00	5.50
5	2023-11-01 00:00:00+05:30	54287355.00	55672825.00	-2.49
6	2023-12-01 00:00:00+05:30	54676485.00	54287355.00	0.72
7	2024-01-01 00:00:00+05:30	56764935.00	54676485.00	3.82
8	2024-02-01 00:00:00+05:30	60036975.00	56764935.00	5.76
9	2024-03-01 00:00:00+05:30	55973020.00	60036975.00	-6.77
10	2024-04-01 00:00:00+05:30	53341470.00	55973020.00	-4.70
11	2024-05-01 00:00:00+05:30	55407955.00	53341470.00	3.87
12	2024-06-01 00:00:00+05:30	55971715.00	55407955.00	1.02
13	2024-07-01 00:00:00+05:30	56617450.00	55971715.00	1.15
14	2024-08-01 00:00:00+05:30	57349170.00	56617450.00	1.29
15	2024-09-01 00:00:00+05:30	53990910.00	57349170.00	-5.86
16	2024-10-01 00:00:00+05:30	56806810.00	53990910.00	5.22
17	2024-11-01 00:00:00+05:30	51502900.00	56806810.00	-9.34
18	2024-12-01 00:00:00+05:30	54406615.00	51502900.00	5.64
19	2025-01-01 00:00:00+05:30	57927360.00	54406615.00	6.47
20	2025-02-01 00:00:00+05:30	51992060.00	57927360.00	-10.25
21	2025-03-01 00:00:00+05:30	53150575.00	51992060.00	2.23
22	2025-04-01 00:00:00+05:30	58511060.00	53150575.00	10.09
23	2025-05-01 00:00:00+05:30	56914875.00	58511060.00	-2.73
24	2025-06-01 00:00:00+05:30	51010265.00	56914875.00	-10.37

```sql
WITH monthly_revenue AS (
  SELECT
    DATE_TRUNC('month', o.orderdate) AS month,
    SUM(oi.totalprice) AS revenue
  FROM orders4 o
  JOIN order_items4 oi ON o.orderid = oi.orderid
  GROUP BY month
)
SELECT
  month,
  revenue,
  LAG(revenue) OVER (ORDER BY month) AS prev_month,
  ROUND(((revenue - LAG(revenue) OVER (ORDER BY month)) /
        NULLIF(LAG(revenue) OVER (ORDER BY month), 0)) * 100, 2) AS growth_percent
FROM monthly_revenue;
```

Insight:

I saw that revenue goes **up and down each month**, which means sales are not steady.

Good months like **April 2025, January 2025, and February 2024** had strong growth.

This could be because of New Year offers, financial year-end sales, or good marketing.

Low months like **Feb 2025, June 2025, and Nov 2024** had drops.

Maybe because there were no offers, less marketing, or low customer interest.

What this tells us:

Look at what worked during the best months and repeat that.

Run special sales or reminders in slow months to keep customers active.

Orders per Country

```sql
CREATE OR REPLACE VIEW v_orders_by_country AS
SELECT
  c.country,
  COUNT(DISTINCT o.orderid) AS total_orders
FROM customers4 c
JOIN orders4 o ON c.customerid = o.customerid
GROUP BY c.country;

select * from v_orders_by_country;
```

	country character varying	total_orders bigint
1	Bangladesh	10049
2	India	9714
3	UAE	10068
4	UK	11264
5	USA	8905

Insight :

India, USA, and UK had the highest orders.

This shows our sales and brand presence are stronger in these regions.

Countries with fewer orders may need better marketing or service.

What this tells us:

Focus more on high-performing countries.

Plan strategies to improve sales in low-order regions.

Average Discount by Payment Method

```
SELECT DISTINCT
  paymentmethod,
  ROUND(AVG(discount) OVER (PARTITION BY paymentmethod), 2) AS avg_discount
FROM orders4 o
JOIN order_items4 oi ON o.orderid = oi.orderid;
```

	paymentmethod character varying	avg_discount numeric
1	Debit Card	0.10
2	UPI	0.10
3	Credit Card	0.10
4	Cash	0.10

Insight:

All payment types — **Debit Card, UPI, Credit Card, and Cash** — got the **same 10% discount**.

This means customers get the same offer no matter how they pay.

What this tells us:

The discount rule is simple and the same for everyone.

But the business can try giving **extra offers** for specific payment methods to:

Encourage people to use UPI or Debit Card

Save on payment processing costs

Overall Avg Days Between Orders (All Customers)

sql

```sql
WITH customer_orders AS (
  SELECT
    customerid,
    orderid,
    orderdate,
    LAG(orderdate) OVER (PARTITION BY customerid ORDER BY orderdate) AS previous_order
  FROM orders
)
SELECT
  customerid,
  orderid,
  orderdate,
  previous_order,
  orderdate - previous_order AS days_between_orders
FROM customer_orders
WHERE previous_order IS NOT NULL;
```

	overall_avg_days_between_orders numeric
1	14.32

Average Time Between Orders – Insight

On average, customers place their next order after **about 14 days**.

This means many customers come back within **two weeks**.

What this tells us:

People seem happy with the products and come back regularly.

The business can use this to grow more sales by:

Sending offers or reminders after 10–12 days

Starting loyalty programs or even subscription plans

Overall Insights & Recommendations

1. Business Performance

- Revenue and order volume are strong.
- Many active customers.
- Most orders are small but frequent.

Recommendation:

Keep offering affordable, fast-moving products and encourage repeat purchases.

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2. Best-Selling Products

- Smartphones, T-shirts, sneakers, laptops, and headphones sell the most.
- Fashion and electronics are top-performing categories.

Recommendation:

Focus marketing and inventory on these items. Try bundle offers or upsell accessories to boost cart value.

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3. Customer Behavior

- Top 5 customers contribute a lot to revenue and come from different countries.
- Male and female customers spend almost equally.

Recommendation:

Start loyalty programs for top customers. Give early access or exclusive offers to keep them coming back.

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4. Country-wise Sales

- USA, India, and UK bring in the most sales.
- Other countries are growing but slower.

Recommendation:

Invest more in ads, delivery, and local offers for top countries. Explore new markets with customized campaigns.

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5. Monthly Sales Trends

- Revenue is steady throughout the year.
- December has the highest sales due to holidays.

Recommendation:

Run major promotions from October to December. Use seasonal campaigns to boost sales.

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6. Campaign Performance

- Google Ads and Email Marketing work best.
- Facebook and Instagram perform okay but can be improved.

Recommendation:

Continue investing in high-performing channels. Test and improve results on social platforms.

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7. Payment Experience

- All payment methods give the same discount (10%).
- No method is being promoted specially.

Recommendation:

Make popular payment options more visible. Try offering extra discounts on low-cost methods like UPI to reduce cart dropouts.